Per-site ΔRMSE from adding mortality — C+T vs C+T+M

Below 0 = mortality reduces the site's RMSE | bars coloured by relative mortality class (site maximum) | repeated CV (3×80 %) | tree cover ≥30 % (101 sites, 361 site-years) · deadtrees v5

ΔRMSE = RMSE(with M) − RMSE(without M)

GBoost (Optuna) · 12r | **GBoost (Optuna) · 24r** | **ghtGBM (Optuna) · 12** | **ghtGBM (Optuna) · 24** | **toGluon (ensemble) · 1** | **toGluon (ensemble) · 2**

GPPsat (µmol m⁻² s⁻¹)

53% improved | 59% improved | 56% improved | 56% improved | 58% improved | 60% improved

NEPmax (µmol m⁻² s⁻¹)

72% improved | 69% improved | 68% improved | 69% improved | 61% improved | 63% improved

ETmax (mm (30 min)⁻¹)

62% improved | 54% improved | 57% improved | 57% improved | 57% improved | 59% improved

WUE (g C kg⁻¹ H₂O)

50% improved | 54% improved | 52% improved | 50% improved | 50% improved | 56% improved

Sites (sorted by ΔRMSE within each panel)

Relative mortality
site maximum, mean of 100-500 m buffers
(breaks: mean ± 0.5 SD)

- Low (< 10.6 %, 41 sites)
- Mid (10.6–24.0 %, 38 sites)
- High (> 24.0 %, 22 sites)